

Article

The Vortex-Induced Vibration Characteristics of the Water-Conveying Truss Pipeline Cable-Stayed Bridge

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Abstract

This study investigated the vortex-induced vibration (VIV) characteristics of a proposed water-conveying truss pipeline cable-stayed bridge through wind tunnel tests. The experimental results indicated that both vertical bending and torsional VIV responses decreased as the wind attack angle increased. The vertical bending VIV behavior of the bridge was significantly influenced by the lateral spacing and relative height of the pipelines. Adjustments to these geometric parameters markedly affected the structural VIV response. Furthermore, computational fluid dynamics (CFD) was employed to analyze the flow field around the truss pipeline bridge. The results revealed that changes in the lateral spacing and relative height of the pipelines primarily altered the VIV performance by modifying vorticity distribution, separation point position, and other critical flow field parameters around the truss section. These findings underscore the importance of considering the effects of geometric parameters on VIV during the design of the truss section in pipeline bridges.

Keywords: bridge engineering; vortex-induced vibration; CFD; truss pipeline bridge; lateral spacing of pipelines; relative height of pipelines



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1. Introduction

With the continuous development of the global economy and accelerated industrialization, the demand for energy continues to grow steadily. In the current era of increasingly scarce resources, selecting energy-efficient transportation methods is of crucial importance. For resources such as oil, water, and natural gas, pipeline transportation not only offers lower costs and reduced loss but also ensures continuous, stable, and high-capacity delivery. Thus, for transporting such critical resources, pipeline transportation serves as a highly effective solution. Common structural forms of pipeline bridges in such systems include cable-stayed, suspension, catenary, and suspended-cable pipeline bridges.

Owing to their characteristically low width-to-height ratio, the installation of pipelines can significantly modify the aerodynamic behavior of the truss section. This alteration may enhance susceptibility to vortex-induced vibration (VIV), thereby potentially jeopardizing the structural safety and operational reliability of the pipelines. In response, a series of studies have been carried out to investigate the underlying mechanisms of VIV and explore effective countermeasures in truss pipeline bridges. Li et al. [1] investigated a water supply pipeline cable-stayed bridge, analyzing its static characteristics, dynamic characteristics, and stability through finite element numerical simulation, demonstrating that, compared with bridges used for transportation, dedicated pipeline bridge exhibited higher load-bearing capacity, necessitating increased focus on structural safety. Tian [2]

conducted research on the VIV characteristics of truss pipeline bridges through wind tunnel test and computational fluid dynamics (CFD) numerical simulations. It was found that, compared with the empty pipeline state, filling the pipeline with water can significantly improve the VIV performance; at the same time, it was revealed that the VIV characteristics of the truss pipeline bridge are significantly associated with the geometric parameters of the pipelines. Zhou [3] conducted a wind tunnel test study on the VIV performance and anti-vortex measures of the double-layer truss girder section at a regular scale (1:60), and found that there were significant vertical and torsional VIV under $+3^\circ$ and 0° wind attack angles, and the amplitude of VIV response of the main girder gradually decreases as the wind attack angle decreased. Yu et al. [4] conducted research on the pipeline bridge section under VIV and established a calculation method for fuzzy reliability under VIV and determined the corresponding calculation and judgment criteria. Wang et al. [5] conducted an analysis and research on the wind-induced response of large-span pipeline bridges and pointed out that the lateral wind acting on the pipeline would generate vortices behind it, causing the pipeline to undergo VIV. Yao et al. [6,7] found that the double-layer truss bridge exhibited significant VIV under specific wind attack angles. However, VIV can be effectively suppressed by installing fairings on the upper chords and inverted-L-shaped guide plates on the lower chords. Additionally, arranging guide plates on the web members and lower chords also proves effective in mitigating VIV. Fang et al. [8] conducted in-depth research on the VIV of double-layer truss bridges, observing significant VIV responses in the wind tunnel test. Numerical simulation further revealed the presence of different frequencies of vortices in the wake zone, and the vertical VIV might be driven by the vortices behind or above the upper bridge deck, while the torsional VIV might be attributed to the vortices behind or below the lower bridge deck. Yu [9] studied the wind resistance and icing characteristics of large-span pipeline bridges and found that the three-component force coefficient of the pipeline under icing was significantly affected by the Reynolds number, surface roughness, icing shape, and turbulence degree, and the asymmetry of the cross-section increased. Hu et al. [10] found that pipeline bridges are susceptible to wind-induced vibration, and the wind-induced response increases with the service life. Li [11] set up a pipeline in a model of a large-span flat steel box girder cable-stayed bridge and found that, when the pipeline was placed inside the wind nozzle of main girder, a significant VIV phenomenon occurred under the $+3^\circ$ wind attack angle condition. Ru [12] studied the structural response of the pipelines when the average wind speed was 2.67 m/s, and analyzed that the cables spanning the pipeline bridge were sensitive to the wind load and prone to generating VIV. At the same time, it was found that the flexible cables had a certain inhibitory effect on the resonance of the pipeline bridge under certain conditions. On railway bridges, the aerodynamic shape was similar to that of pipeline bridges. Yu et al. [13] conducted wind tunnel tests and found that the lateral spacing between pipelines had a relatively small impact on the average aerodynamic force coefficient, while optimizing the vertical spacing between pipelines could significantly reduce the average aerodynamic force coefficient. Yang et al. [14] studied the VIV performance of a streamlined pipeline box girder bridge. In the wind tunnel test, severe vertical VIV was observed in the pipeline bridge within a lower wind speed range. Based on the flow pattern around the cross-sectional model, the pipeline completely separated the flow on the bottom plate surface, causing periodic shedding vortices in the wake zone and inducing bottom plate vibration. Wang et al. [15] discussed the influence of structural parameters and the cross-sectional shape of the main girder on the wind resistance performance of the suspension pipeline bridge, and believed that changing the cross-sectional shape of the main girder could effectively prevent the formation of VIV. Li [16] further demonstrated that, in single-layer pipeline bridge models, the drag coefficient increased almost linearly with pipeline diameter. For

double-layer pipeline bridge models, when the diameter of the lower pipeline was fixed, increases in the upper pipeline diameter led to corresponding increases in both resistance coefficient and moment coefficient under different wind attack angles that were consistent.

In conclusion, the distinctive aerodynamic profile of pipeline bridges makes it essential to study how pipeline parameters influence their wind-induced vibration characteristics. It can be observed that no studies have yet specifically investigated the influence of pipeline parameters on the VIV characteristics of truss pipeline bridges. Therefore, researching the impact of these pipeline parameters on the VIV performance of truss pipeline bridges is highly necessary.

This study examines the influence of pipeline lateral spacing and relative height on the VIV characteristics of a truss pipeline cable-stayed bridge, using both wind tunnel tests and computational fluid dynamics (CFD). The pipelines are modeled in an empty state, based on the assumption that the primary effect of conveyed water is added mass, and that the fluid filling condition does not substantially modify the influence of geometric parameters on VIV. Vorticity evolution is analyzed to elucidate the mechanisms underlying the parametric effects. The findings provide valuable insights for the design and engineering practice of pipeline bridges.

2. Wind Tunnel Tests

2.1. Project Profile

The proposed water-conveying pipeline bridge has a total length of 1230 m, with a 490 m main bridge designed as a four-tower cable-stayed structure employing a steel truss system. Figure 1 presents the elevation and standard cross-section of the main girder.

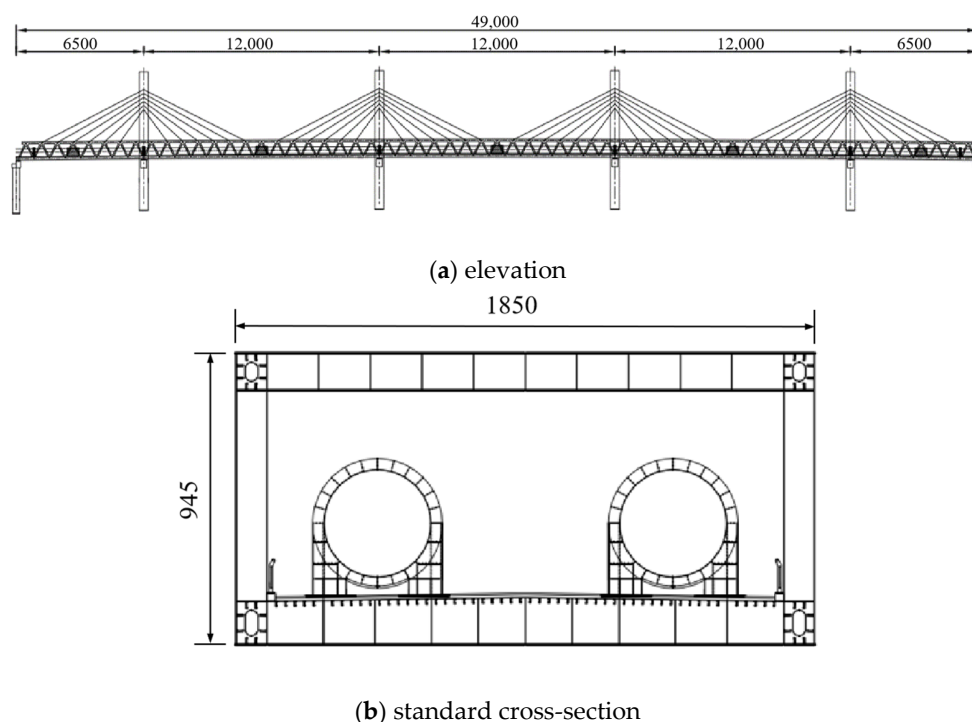


Figure 1. Layout of pipeline bridge (unit: cm).

2.2. Sectional Model

The wind tunnel tests were carried out in the CA-01 wind tunnel at Chang'an University using a sectional model scaled at 1:32. Figure 2 shows the sectional model of the pipeline bridge mounted on the support frame inside the wind tunnel, along with the cross-sectional profile of the model. The upper and lower chords of the model were made

of aluminum tubes to ensure structural rigidity, while the outer contour was fabricated from ABS plates to accurately represent the bridge geometry while maintaining light weight and sufficient stiffness. The structural stiffness was provided by springs. In addition, elliptical wooden plates were employed as binary end plates to effectively mitigate the influence of end effects on wind pressure distribution, meeting both functional requirements and rigidity criteria. The data acquisition system consisted of a laser displacement sensor, a DASPV11 data acquisition, and a computer. Relevant design parameters are summarized in Table 1. Note: f_b and f_t denote the vertical bending and torsional frequencies, respectively; M and J represent the equivalent mass and mass moment of inertia per unit length of the girder, respectively; ζ_b and ζ_t are the vertical bending and torsional damping ratios, respectively.

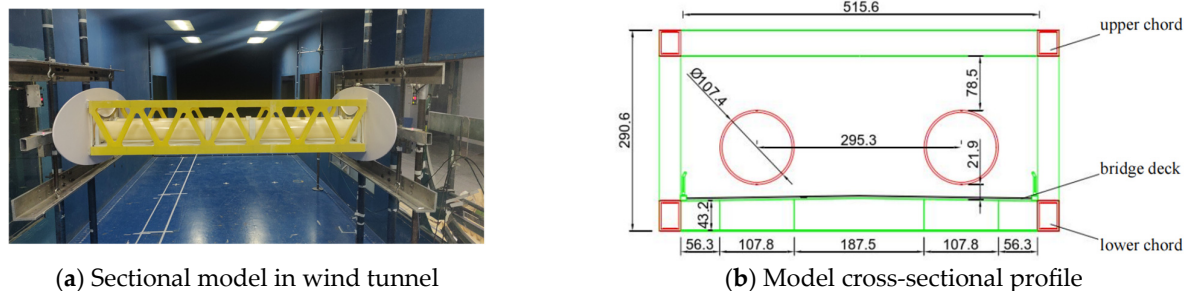


Figure 2. Wind tunnel test sectional model (unit: mm).

Table 1. Model design parameter.

Parameters	Units	Prototype	Scale Ratios	Test Value
M	kg/m	21,000	1/322	20.507
J	kg m ² /m	775,000	1/324	0.739
f_b	Hz	0.788	32/3.2	7.880
f_t	Hz	1.781	32/3.9	14.613
ζ_b	%	1	/	0.5
ζ_t	%	1	/	0.5

2.3. Experimental Results

The VIV test of the bridge was conducted in a uniform flow field with incoming wind attack angles set at -3° , 0° , and $+3^\circ$. The wind speed and vibration amplitudes were scaled according to the actual bridge parameters. Figure 3 presents the VIV amplitudes of the truss pipeline bridge under these wind attack angles.

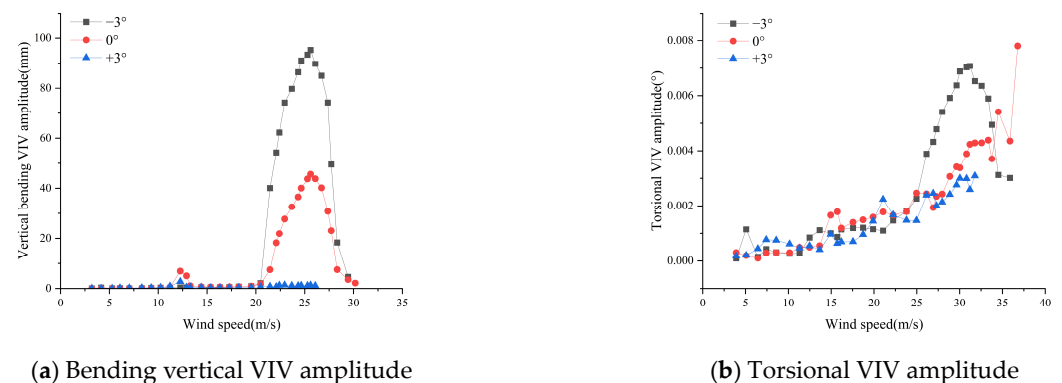


Figure 3. Pipeline bridge VIV amplitude under different wind attack angles.

As shown in Figure 3, the amplitude of vertical bending VIV of the truss pipeline bridge decreases progressively as the wind attack angle increases. At a wind attack angle

of -3° , the vertical bending VIV amplitude reached 95.28 mm. When the angle increased to 0° , the amplitude decreased significantly to 45.63 mm, representing a reduction of approximately 50%. As the wind attack angle further increased to $+3^\circ$, the vertical bending VIV amplitude diminished nearly to zero. The VIV behavior of the truss pipeline bridge is predominantly characterized by vertical vibration, with torsional VIV being negligible.

3. Influence of Pipeline Geometric Parameters on the Characteristics of Vertical Bending VIV

3.1. Experimental Conditions

This section examines the influence of pipeline lateral spacing and relative height on the VIV performance of the pipeline bridge. The definitions of the experimental parameters are illustrated in Figure 4. The girder height and width are denoted as H and B , respectively. The center-to-center distance between the two pipelines is designated as X . The vertical distance from the lower edge of the pipeline to the bridge deck is defined as Z_1 , and the vertical distance from the bridge deck to the lower edge of the upper chord member is specified as Z_2 .

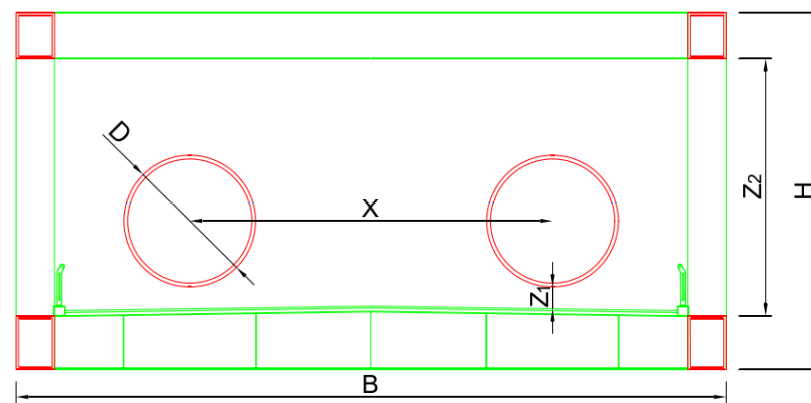


Figure 4. Experimental parameters.

Furthermore, the amplitude corresponding to the peak point of the VIV interval is defined as A_c ; the VIV onset wind speed is U_a ; and the length of the lock-in interval is L_S . Figure 5 is a schematic diagram of the VIV lock-in interval, where a, b, and c represent the onset point, ending point, and peak point, respectively.

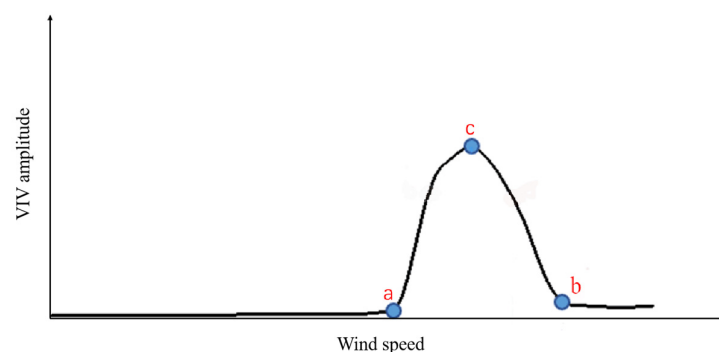


Figure 5. Lock-in interval.

To facilitate the comparison and analysis of the test results, the geometrical parameters are designed to be dimensionless. The parameter $\lambda = X/B$ is defined as the lateral spacing of the pipelines, and the parameter $\gamma = Z_1/Z_2$ is defined as the relative height of the pipelines. The layout of the test conditions is shown in Table 2.

Table 2. Test conditions.

Dimensionless Parameters	X (mm)	B (mm)	$\lambda = X/B$
λ	202.3	578	0.35
	231.2		0.4
	260.1		0.45
	289		0.5
	318		0.55
	Z_1 (mm)	Z_2 (mm)	$\gamma = Z_1/Z_2$
γ	/	/	0.25
			0.375
			0.5
			0.75

3.2. Influence of Lateral Spacing of Pipelines on VIV

3.2.1. VIV Peak Amplitude

The dimensionless peak amplitude of VIV is defined as the ratio of the amplitude at the peak point to the height of the girder (A_c/H). Figure 6 shows the curve depicting the influence of the lateral spacing of the pipelines λ on the peak amplitude.

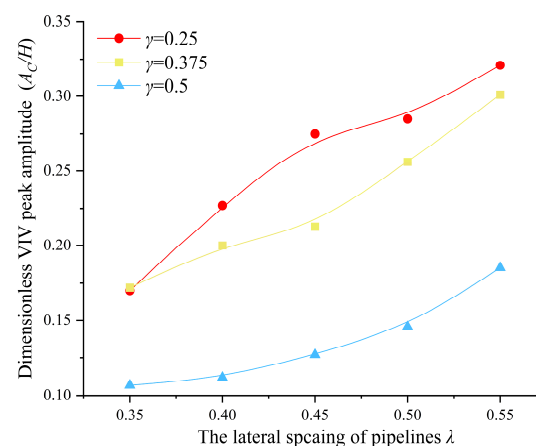


Figure 6. The influence of lateral spacing of pipelines on VIV peak amplitude.

As shown in Figure 6, the peak amplitude of VIV exhibits a significant increasing trend as the lateral spacing ratio λ of the pipelines gradually increases. When the lateral spacing is small, the peak amplitude rises with a relatively mild growth rate. However, as λ approaches the critical value of half the main girder width, the growth trend undergoes a pronounced shift, displaying a sharp, rapid increase. These results indicate that a pipeline spacing close to half the girder width is unfavorable for mitigating VIV.

3.2.2. VIV Lock-In Interval Length

Figure 7 shows the curve depicting the influence of the lateral spacing of the pipelines λ on the length of the VIV lock-in interval.

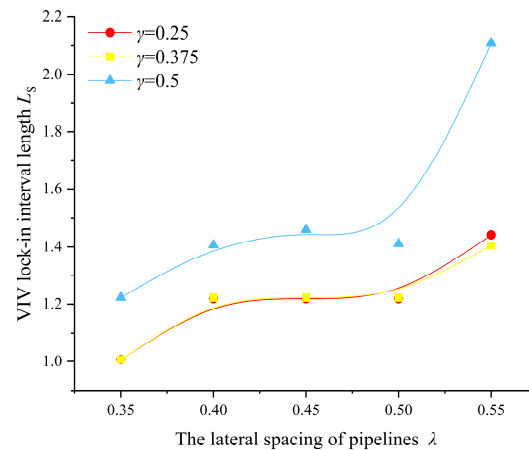


Figure 7. The influence of lateral spacing of pipelines on VIV lock-in interval length.

As shown in Figure 7, when the lateral spacing of the pipelines λ is relatively small (less than half the main girder width), the lock-in interval length L_s remains largely stable. However, once λ exceeds 0.5, the lock-in interval length expands abruptly. This suggests the existence of a critical value—approximately equal to half the girder width—beyond which further increases in λ lead to a pronounced extension of the VIV lock-in interval length.

3.2.3. VIV Onset Wind Speed

The dimensionless onset wind speed of VIV was defined as the ratio of the VIV onset wind speed to the product of the girder height and the fundamental vertical bending frequency ($U_a/f_b H$). Figure 8 illustrates the curve depicting the influence of the lateral spacing of the pipelines λ on the VIV onset wind speed.

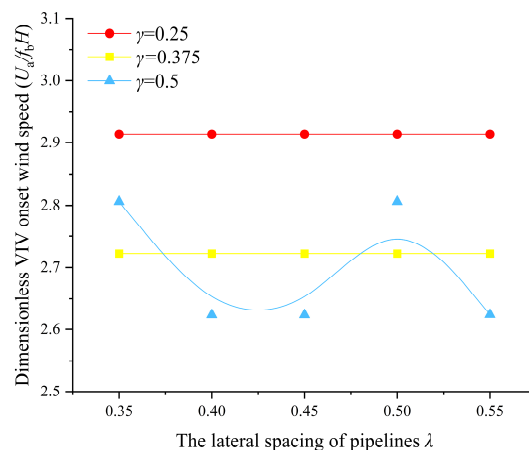


Figure 8. The influence of lateral spacing of pipelines on VIV onset wind speed.

As shown in Figure 8, the influence of the lateral spacing of pipelines λ on the onset wind speed of VIV does not exhibit a consistent increasing or decreasing trend. For $\gamma = 0.25$ and $\gamma = 0.375$, the VIV onset wind speed remains largely unchanged as λ increases, indicating relatively low sensitivity to variations in lateral spacing.

In contrast, at $\gamma = 0.5$, the behavior differs noticeably: the VIV onset wind speed varies in a cosine-like pattern with increasing λ , demonstrating significantly higher sensitivity to lateral spacing changes. These results suggest that the sensitivity of the onset wind speed to λ varies with the relative height ratio γ , though the overall influence of λ on VIV onset remains limited.

3.3. Influence of Relative Height of Pipelines on VIV

3.3.1. VIV Peak Amplitude

Figure 9 shows the curve depicting the influence of the relative height of the pipelines γ on the VIV peak amplitude.

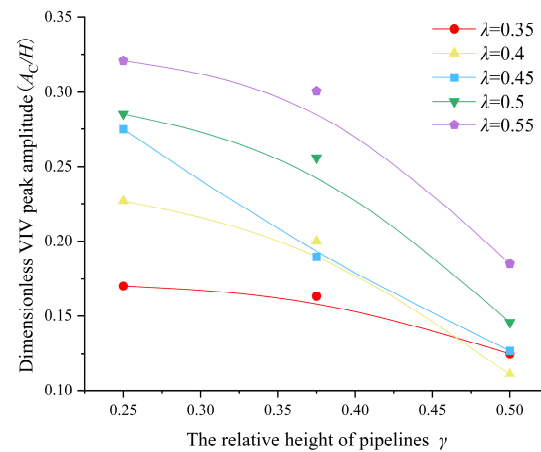


Figure 9. The influence of relative height of pipelines on VIV peak amplitude.

As shown in Figure 9, the VIV peak amplitude exhibits a clear decreasing trend as the relative height of pipelines γ increases. When the lateral spacing of pipelines λ is small, variations in γ have limited influence on the peak amplitude. In contrast, at larger values of λ , the effect of γ becomes considerably more pronounced. A comparison across different lateral spacing conditions reveals that increasing γ consistently reduces the VIV peak amplitude.

3.3.2. VIV Lock-In Interval Length

Figure 10 shows the curve depicting the influence of the relative height γ of the pipelines on the length of the VIV lock-in interval.

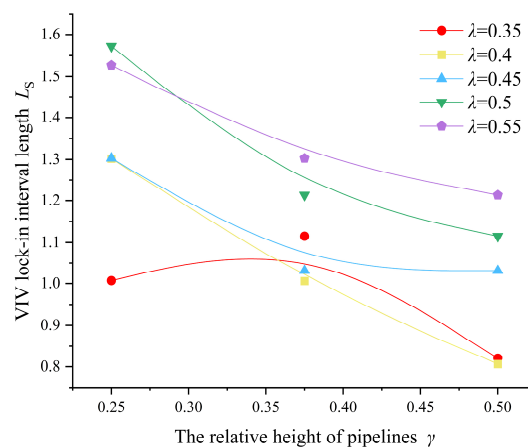


Figure 10. The influence of relative height of pipelines on VIV lock-in interval length.

As shown in Figure 10, the length of the VIV lock-in interval, L_s , generally decreases as the relative height of pipelines γ increases. At $\lambda = 0.35$, a critical relative height of $\gamma = 0.375$ is observed, beyond which the lock-in interval length decreases sharply. For other lateral spacings, the rate of decrease in L_s gradually slows down with increasing γ .

3.3.3. VIV Onset Wind Speed

Figure 11 shows the curve depicting the influence of the relative height of the pipelines γ on the VIV onset wind speed.

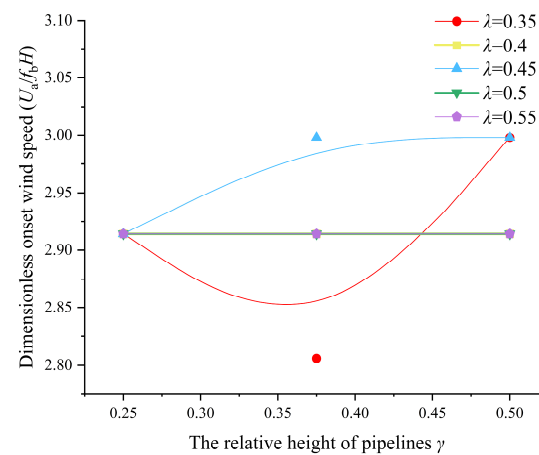


Figure 11. The influence of relative height of pipelines on VIV onset wind speed.

As shown in Figure 11, the sensitivity of the VIV onset wind speed U_a to changes in the relative height of pipelines γ varies with the lateral spacing of pipelines λ . When $\lambda = 0.35$, U_a exhibits strong sensitivity to γ , initially decreasing and then increasing as γ rises. At $\lambda = 0.45$, U_a remains highly responsive to variations in γ , first increasing before stabilizing with further increases in relative height. In contrast, for $\lambda = 0.4, 0.5$, and 0.55 , the VIV onset wind speed shows negligible change, indicating minimal sensitivity to variations in γ .

4. CFD Calculation

4.1. Two-Dimensional Numerical Model Layout

4.1.1. Computational Domain

The dimensions of the model in the two-dimensional computational domain match those of the actual sectional model used in the wind tunnel tests. The layout and sizing of the computational domain are illustrated in Figure 12.

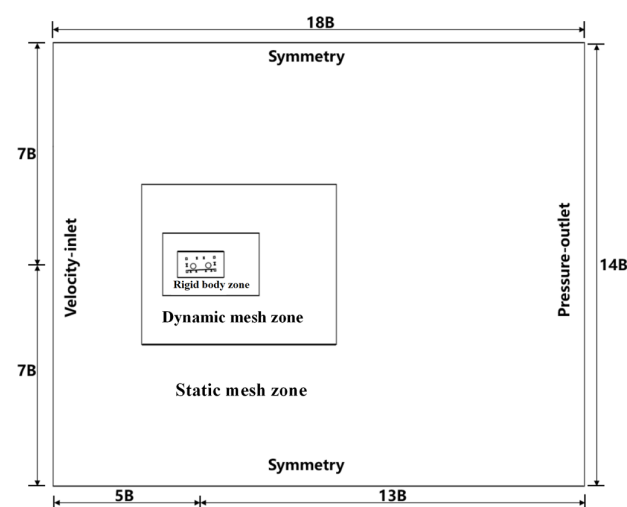


Figure 12. Two-dimensional computing domain.

The left boundary of the computational domain is defined as the velocity inlet with a uniform incoming flow at 5% turbulence intensity. The right boundary is set as the pressure outlet, while the upper and lower boundaries are defined as symmetry.

4.1.2. Mesh Generation

The static mesh zone and the rigid body zone are discretized using structured grids, whereas the dynamic mesh zone utilizes an unstructured grid. The Shear Stress Transport (SST) $k-\omega$ turbulence model is adopted. The first layer of the boundary mesh has a height of 1×10^{-5} m, satisfying $y^+ < 1$. Three mesh resolutions were considered for grid independence verification, resulting in a final total cell count of approximately 270,000.

Numerical simulations were carried out using the SIMPLE algorithm in ANSYS Fluent 2023 R1, with a residual convergence criterion of 1×10^{-5} and a time step size of 0.001 s. The total simulation time was set to 30 s. Figure 13 illustrates the mesh configuration, where subfigures (a) to (d) present a progressively zoomed-in view within the same image.

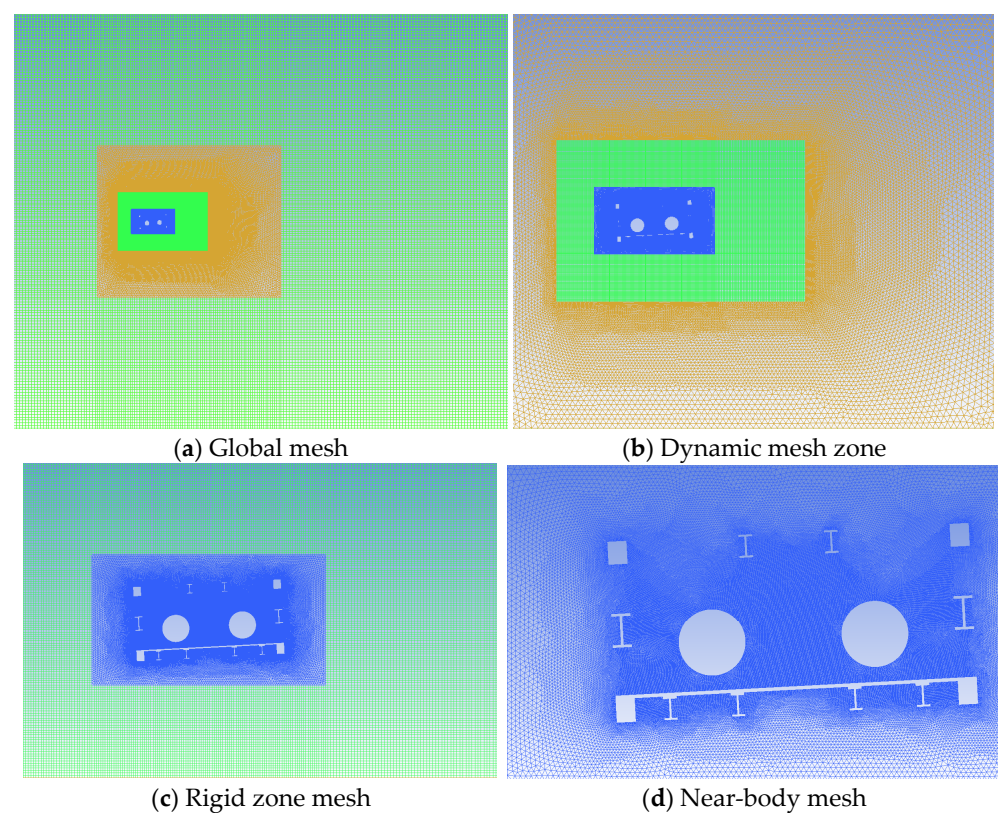


Figure 13. Two-dimensional model mesh generation.

4.1.3. Mesh Validation

Since the pipeline bridge experienced vertical vortex-induced vibration, this study focuses specifically on the lift coefficient. The inflow wind speed was consistent with the test conditions, and force measurement simulations were conducted at wind attack angles ranging from -10° to 10° with 1° intervals, totaling 21 angles. A comparison between the CFD results and test findings is shown in Figure 14.

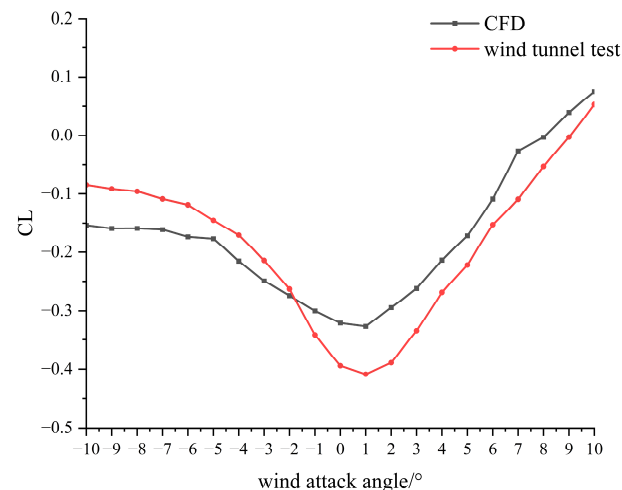


Figure 14. Comparison of CL results from CFD and wind tunnel test.

As shown in Figure 14, the lift coefficient obtained from the CFD simulations demonstrates good agreement with the wind tunnel test results, confirming that the numerical approach employed in this study accurately captures the VIV behavior of the truss pipeline bridge.

4.2. Influence of the Lateral Spacing of Pipelines on the Flow Field

Figure 15 shows the evolution of the flow field of the truss pipeline bridge over one complete cycle after VIV reaches steady-state amplitudes, under different lateral spacings of the pipelines λ . To facilitate comparison, vortices in different regions are marked in the vorticity contour plots.

Figure 15a,b show airflow separation at the windward web member. The resulting vortices G-CCW-1 and G-CW-1 are significantly influenced by the lateral spacing of pipelines λ . As λ increases, both the development distance and vorticity magnitude of these vortices intensify. Specifically, the shed vortex G-CCW-1 impacts the upper-left region of the windward pipeline. With increasing λ , the energy of both G-CCW-1 and G-CW-1 correspondingly amplifies.

The vortex shedding behavior at the windward pipeline tail exhibits marked variations with increasing λ . Observations from Figure 15c,d reveal that vortices G-CCW-2 and G-CW-2 reach peak intensity at the bridge deck centerline. As λ grows, G-CCW-2 gradually weakens, and the energy of leeward-side G-CW-2 also diminishes. The leeward wake vortices, affected by web members, show changes in energy and attack angle under different λ . Comparing Figure 15g,h, the angle between the wake vortex Q-CW-1 and the leeward lower chord increases with λ , while the vortex structure Q-CCW-1 near the leeward lower chord becomes disrupted and eventually sheds due to Q-CW-1's influence. The enhanced energy and altered angle of Q-CW-1 delay the breakdown of Q-CCW-1, thereby prolonging its energy transfer duration to the structure. In contrast, vortices shed from the upper chord exhibit smaller energy and scale, thus exerting negligible effects on the VIV performance and warranting no further analysis.

With increasing λ , the energy of windward vortices G-CCW-1 and G-CW-1 intensifies, amplifying their impact forces on the windward pipe surface and elevating local vortex energy. Concurrently, the energy of leeward-impacting vortex G-CW-2 progressively decreases, whereas Q-CW-1 gains both energy and angular deflection. This delays the disintegration of Q-CCW-1, extending its interaction period with the structure. These differential energy levels and spatial distributions of key vortices are inferred as primary contributors to the amplified VIV amplitudes.

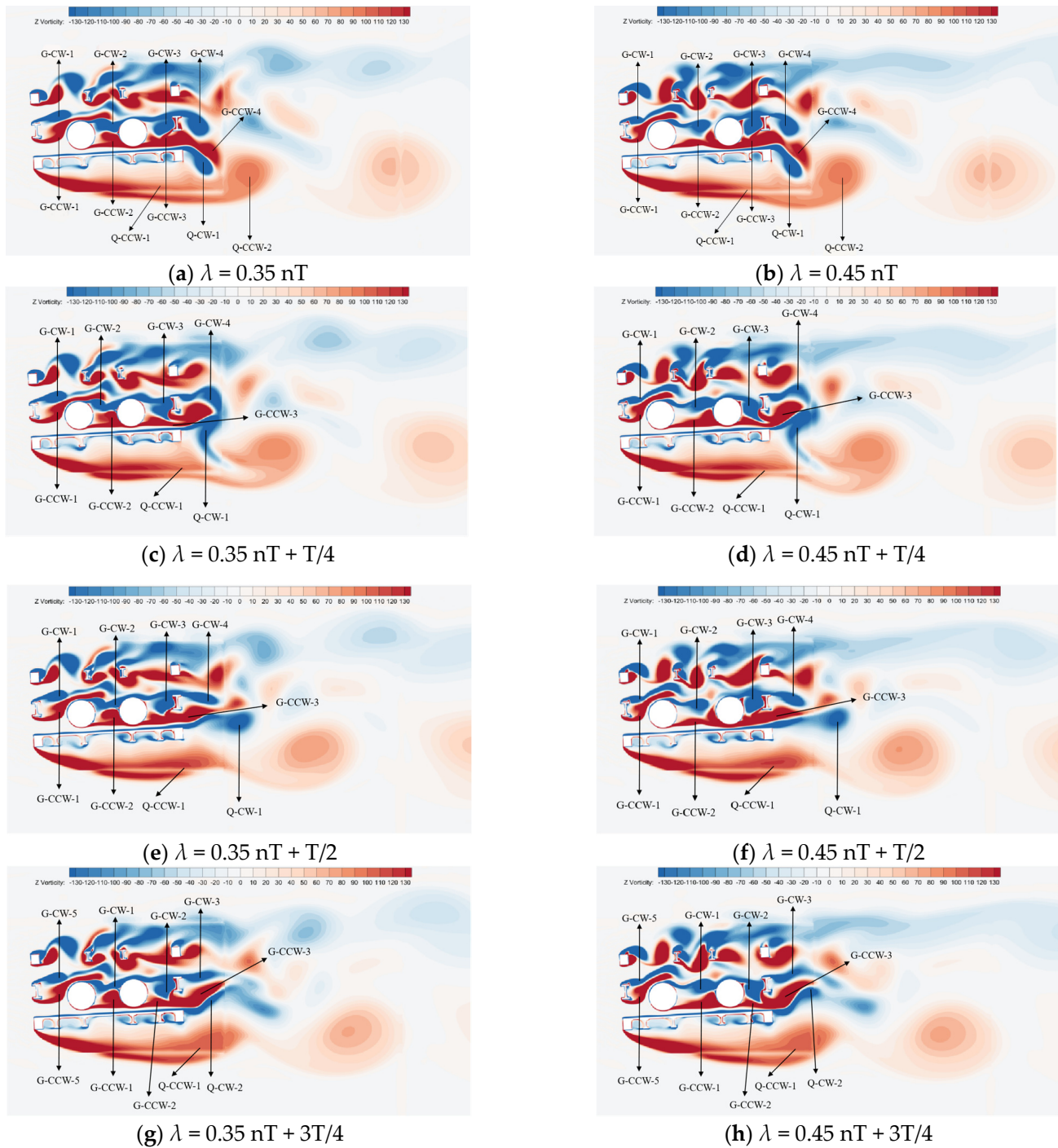


Figure 15. Evolution of flow field in one cycle with different lateral spacings of pipelines.

4.3. Influence of the Relative Height of Pipelines on the Flow Field

Figure 16 shows the evolution of the flow field of the truss pipeline bridge over one complete cycle after VIV reaches steady-state amplitudes, under different relative heights γ . To facilitate comparison, vortices in different regions are marked in the vorticity contour plots.

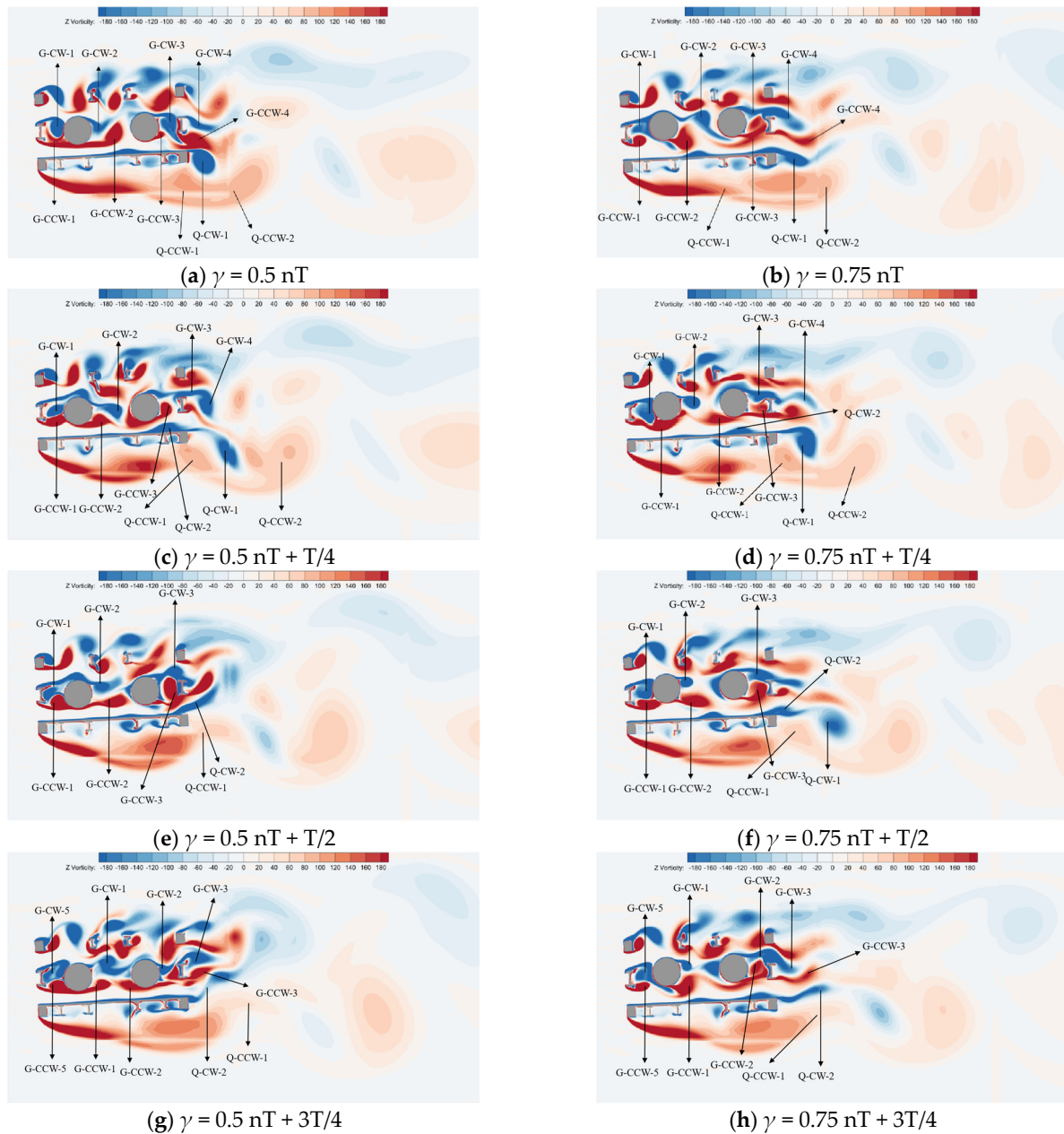


Figure 16. Evolution of flow field in one cycle with different relative heights of pipelines.

Figure 16a reveals that airflow separation occurs near the windward web member, with the wake trajectory and intensity being significantly influenced by the relative height pipelines γ . At lower γ values, vortices G–CW–1 and G–CCW–1 impact the upper-left and lower-left regions of the windward pipeline, respectively, subsequently splitting into two parts: one remains between the windward web member and pipeline, while the other adheres to the pipeline surface and moves along it. The adhered G–CW–1 and G–CCW–1 undergo secondary separation at the upper-right and lower-right regions of the windward pipe, respectively.

At higher γ values, as shown in Figure 16b, the vortices impacting the windward pipeline are predominantly clockwise (CW), with impingement points located on the pipeline's left side. A minor portion consists of counterclockwise (CCW) vortices shed from the inclined web member wake, impacting the lower section of the model. Compared to low γ conditions, both impingement points shift downward. The impacting CW vortices split

into two components: one moves clockwise along the pipeline, while the CCW component interacts with the existing CCW vortices, promoting their separation.

Figure 16c,d demonstrate that the windward pipeline wake vortices G–CW–2 and G–CCW–2 collide at the leeward pipeline, causing airflow reattachment above and below the pipeline before moving downstream. Due to obstruction by the leeward inclined web member, a portion of the separated flow persistently circulates between the leeward pipeline and web member, forming a sustained vortex. The flow separates at the bridge deck, and the leading edge gains energy under the influence of nearby vortices and ultimately sheds as a CW vortex (Q–CW–1) at the trailing edge. Meanwhile, vortices on the lower deck surface develop from the leading edge and gradually bifurcate into Q–CCW–1 and Q–CCW–2. As seen in Figure 16e,f, Q–CW–1 disrupts Q–CCW–1, splitting it into two vortices and causing premature shedding of the rear vortex.

With increasing γ , the vortex structure around the windward pipeline shifts toward predominantly CW vortices, with slightly lowered impingement points. This enhances the energy and size of vortices beneath the pipeline, amplifying their influence on the deck. Changes in vortex energy and separation–reattachment positions on the pipeline surface increase the negative work performed by the shedding CCW vortices on the deck, consequently reducing the VIV amplitude.

5. Conclusions

Through carefully controlled wind tunnel test and CFD simulation, this study investigated the influence of two specific pipeline geometric parameters on the vertical VIV of a water-conveying truss pipeline cable-stayed bridge. It is important to note that these findings are derived under specific conditions, including a fixed wind angle of attack and a particular structural configuration. The main conclusions are as follows:

1. As the lateral spacing of the pipelines λ increases, both the VIV amplitude and the lock-in interval length increase, while the VIV onset wind speed shows no clear trend. This overall indicates a deterioration in VIV performance. The increase in λ enhances the energy of the detached vortices on the windward side; thus, the impact force on the windward side pipeline also increases, leading to an increase in the energy of the surface vortices on that side, and resulting in a deterioration of the VIV characteristics.
2. As the relative height of the pipelines γ increases, both the VIV amplitude and the lock-in interval length decrease, while the VIV onset wind speed does not exhibit a consistent trend. These changes collectively indicate an improvement in VIV performance. The increase in γ leads to changes in the vortex structure on the windward side of the pipeline, with the impact point moving downward. The energy size of the vortex below the pipeline is strengthened, as well as the changes in the vortex energy on the pipeline surface and the separation point and reattachment point. This results in the emergence of counterclockwise vortices, thereby improving the vortex performance.

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Abbreviations

The following abbreviations are used in this manuscript:

VIV Vortex-induced vibration

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